# Task 3: Login System Comparison Report

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**Student Code vs AI Generated Code**

## Feature Comparison Table

| Feature | Student Code | AI Code |
|---|---|---|
| Structure | Simple | Advanced |
| Readability | Easy | Medium |
| Security | Basic | Good |
| Session Handling | Yes | Yes |

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# Explanation of Comparison

### 1. Structure

The **student-written code uses a simpler structure**, focusing mainly on completing the required features such as registration, login, session creation, and route protection. The files are organized but the logic is straightforward.

The **AI-generated code uses a more advanced structure**, including improved error handling, password hashing, and additional validation. The AI version also follows better backend development practices.

### 2. Readability

The **student code is easier to read** because it uses simple logic and fewer layers of security or validation. This makes it more beginner-friendly.

The **AI-generated code has medium readability** because it includes additional features such as password hashing and structured responses, which slightly increase the complexity of the code.

### 3. Security

The **student version has basic security** because passwords are stored directly in the database without encryption.

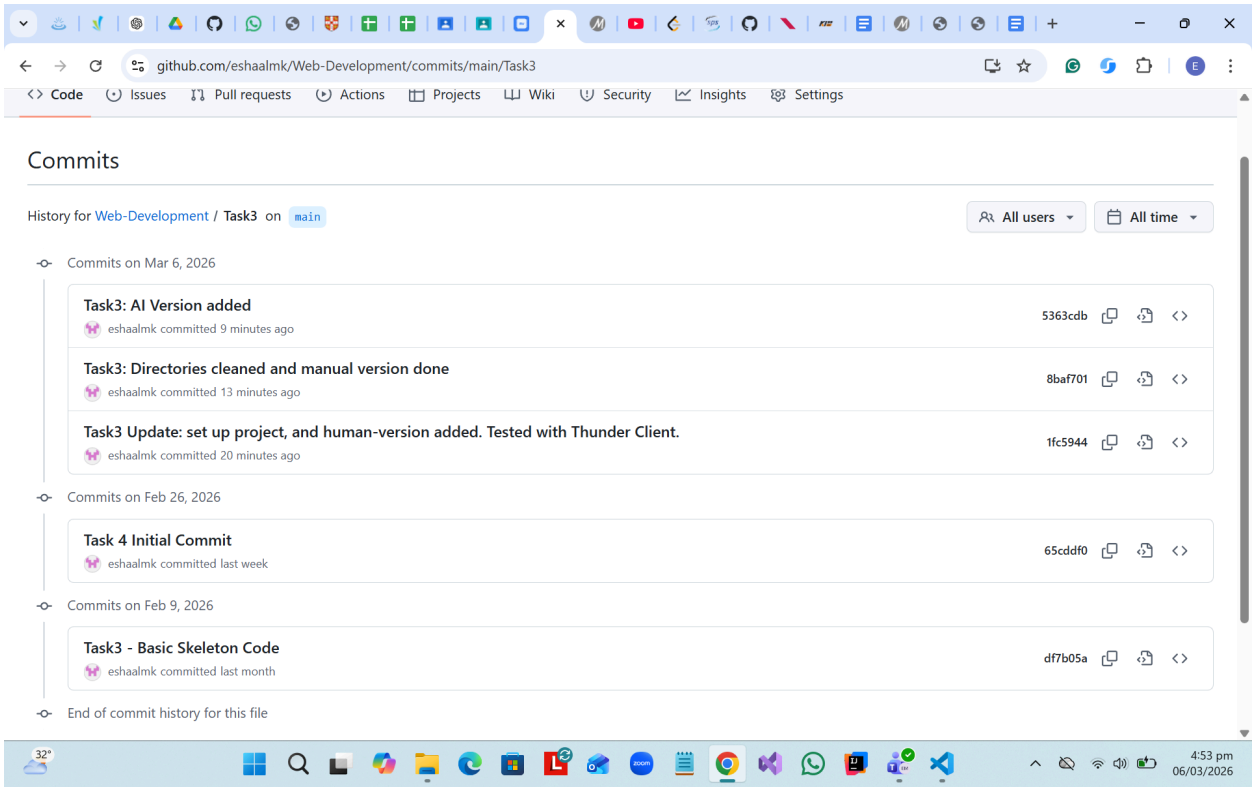
The **AI-generated version provides better security** by hashing passwords using the bcrypt library before storing them in the database. This prevents passwords from being stored in plain text.

### 4. Session Handling

Both the **student code and AI-generated code implement session handling** using the express-session library. Sessions are created after a successful login and are used to protect the dashboard route so that only authenticated users can access it.

# Conclusion

Both implementations successfully fulfill the requirements of the login system assignment, including user registration, login authentication, session handling, protected routes, and logout functionality.

However, the **AI-generated version demonstrates more advanced backend practices**, especially in terms of security and error handling, while the **student-written version focuses on simplicity and basic functionality.**

Github: [https://github.com/eshaalmk/Web-Development/tree/main/Task3](https://github.com/eshaalmk/Web-Development/tree/main/Task3)